Mahdi Asefi

AI Engineer — LLM Systems, Computer Vision & Generative AI

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Open to new roles in AI engineering — LLM & agent systems, computer vision, and generative AI.

SUMMARY

AI Engineer with 4+ years building production AI systems across LLM agents, computer vision, and generative AI. Shipped 3 AI products and 7+ models to production at PiRun (Drivee.ai); now sole engineer on an end-to-end generative video platform at Metadeep. M.Sc. in Artificial Intelligence (GPA 18.89/20), thesis in progress on face forgery detection.

TECHNICAL SKILLS

Languages & Core: Python · SQL · Git · Linux · Docker

Deep Learning & Vision: PyTorch · Hugging Face Transformers · T5 / mT5 · scikit-learn · TensorFlow · OpenCV · YOLO · ViT · CNNs · GANs · Matplotlib

LLM & Agent Systems: OpenAI API · LangChain · LangGraph · RAG · Qdrant · FAISS · prompt engineering · tool & function calling · multi-agent systems · LLM evaluation & fine-tuning · Langfuse · Redis-backed conversational memory · ElevenLabs (TTS / voice cloning)

Generative Media & Video: ComfyUI · ComfyUI API · ControlNet · IP-Adapter · VAE · diffusion models · Vision LLM / VLM · FFmpeg · Vast.ai (on-demand cloud GPU) · image & video prompt engineering

MLOps & Infrastructure: model deployment & serving · ONNX · MLflow · CI/CD · async & background job queues

Domain & Research: deepfake & face forgery detection · face detection & identification · GANs for medical imaging · synthetic data generation · Persian NLP · question answering systems · multimodal / VLM pipelines · image processing · data cleaning & augmentation · hyperparameter tuning

EXPERIENCE

Full-Stack AI Engineer — Metadeep

Karaj, Alborz, Iran · Hybrid · Feb 2026 – Present

- Architected an end-to-end generative video platform as sole engineer: frontend, backend, AI pipeline.
- Designed a LangGraph multi-agent system for automated video scripting with human-in-the-loop review.
- Integrated and benchmarked **5+ generative video/image models** via the ComfyUI API (ControlNet, VAE).
- Built a Vision LLM engine extracting visual properties into creative briefs and shot-level prompts.
- Architected on-demand GPU provisioning on Vast.ai, spinning compute up per job, not idle instances.
- Integrated a Persian/Farsi voice-over pipeline (voice cloning and TTS) with RTL subtitle rendering.

AI Engineer — Tetronix

Remote (Canada) · Sep 2025 – Dec 2025

- Built and deployed an AI chatbot on FastAPI using the OpenAI API, with Langfuse for observability.
- Implemented Redis short-term and summarised long-term memory, **cutting token usage by ~30%**.
- Integrated real-time ElevenLabs voice interaction, sharing conversational state via Redis.
- Built a web-crawling pipeline aggregating car-marketplace data into MongoDB for related-car search.

AI Engineer — PiRun (Drivee.ai)

Remote (US) · Mar 2023 – Jul 2025

- Joined early; contributed to **three production AI products**: Drivee AI, Drivee Sales Agent, and HeyGent.
- Took **7+ models to production**: segmentation, YOLO detection, bokeh, CGAN shadows and reflections.
- Built multi-agent systems for sales, booking, and engagement, **handling hundreds of leads per month**.
- Applied CGANs for realistic shadows and reflections, optimised for fast CPU inference via ONNX.
- Fine-tuned LLMs and integrated chatbot systems with CRM, DMS, and scheduling tools.

NLP Engineer Intern — Commercialization and Innovation Center, University of Kashan

Isfahan, Iran · Apr – Jun 2022

- Fine-tuned a multilingual mT5 question-answering system with Hugging Face Transformers.

Computer Vision Engineer (Part-time) — Sina Teb

Isfahan, Iran · Mar – Oct 2021

- Fine-tuned YOLOv5 for long-distance UAV car detection, then deployed a real-time tracking system.

Teaching Assistant — Fundamentals of Computer and Programming, University of Kashan

Kashan, Isfahan, Iran · Feb – May 2019

- Designed assignments and taught C++ fundamentals to ~10 students, including debugging and grading.

RESEARCH EXPERIENCE

Deepfake and Face Forgery Detection — Research Assistant

Supervisor: Prof. Azadeh Mansouri (Kharazmi University) · 2024 – Present

- Reviewed five deepfake-detection papers, presenting findings and methodologies to faculty and team.
- Trained models from two reviewed papers and combined their strengths for improved deepfake detection.
- M.Sc. thesis work; RFAM (AAAI 2021) combined with CORE (CVPRW 2022) published as open source.

GANs in Radiogenomics — Research Assistant

Supervisor: Prof. Qian Liu · Jan – Mar 2023

- Studied GAN-based synthetic data generation for breast cancer detection using MRI and genomics.
- Investigated fusing genomic data with MRI feature maps into a unified generation network.

Face Detection and Clustering in News Imagery — Research Assistant

Supervisor: Dr. Pooria Lajevardy · March 2022

- Summarised two papers on face detection in news images, critiquing and improving their methods.

EDUCATION

M.Sc. in Artificial Intelligence — Azad University of Tehran Central Branch

Tehran, Iran · 2023 – Present (Expected 2026) · GPA 18.89/20

Thesis: Deepfake and face forgery detection. Supervisor: Prof. Reyhaneh Tati (Azad University) · Consulting Supervisor: Prof. Azadeh Mansouri (Kharazmi University)

B.Sc. in Computer Science — University of Kashan

Isfahan, Iran · 2018 – Jan 2023 · GPA 16.64/20

Awarded a stipend for exceptional performance on the undergraduate research project "Persian Paraphrasing". Supervisor: Prof. Mahsa Soheil Shamaee

SELECTED PROJECTS

Combined Face Forgery Detection Method (RFAM × CORE)

github.com/mahdiAsefi20/RFAMxCORE

- Unified training pipeline coupling RFAM's frequency-aware cues with CORE's consistency on Xception.
- Trained and evaluated on FaceForensics++, with a dedicated path for FF++ low-quality splits.

RFAM — Local Relation Learning for Face Forgery Detection (AAAI 2021)

github.com/mahdiAsefi20/RFAM

- From-scratch reimplementation of the paper's RGB-frequency attention and patch-similarity modules.
- Built the full dataset pipeline for FaceForensics++, DFDC, DFFD, and Celeb-DF.

BERT from Scratch

github.com/mahdiAsefi20/miniBERT

- Implemented BERT in PyTorch — tokenizer, embeddings, encoder stack, and training, pretrained via MLM.

Persian Paraphrasing

Undergraduate Research Project, University of Kashan · Jul 2022 – Nov 2022 · github.com/mahdiAsefi20/Persian-Paraphraser

- Curated a Persian paraphrase corpus from ParaNMT-50M, chunked to fit Colab memory limits.
- Fine-tuned mT5 for Persian paraphrasing and co-authored a 14-page paper with Prof. Shamaee.

LANGUAGES

Persian — Native

English — Professional working proficiency (TOEFL iBT 91/120: Reading 24 · Listening 24 · Speaking 20 · Writing 23)